Worksheet 13 · Inequalities on Graphs IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Draw straight-line graphs to show inequalities • Shade or identify regions that satisfy several inequalities • Find integer coordinates inside a region		
Grade: / 41	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Inequalities on Graphs*

Question 1 (4 marks)

(a) Write down the integer values of x that satisfy the inequality $-2 < x \leqslant 4$

(2)

The region **R**, shown shaded in the diagram, is bounded by three straight lines.

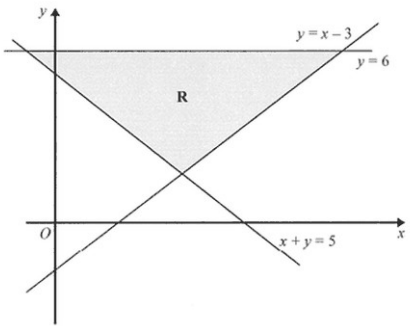


Diagram **NOT** accurately drawn

(b) Write down the three inequalities that define the region **R**.

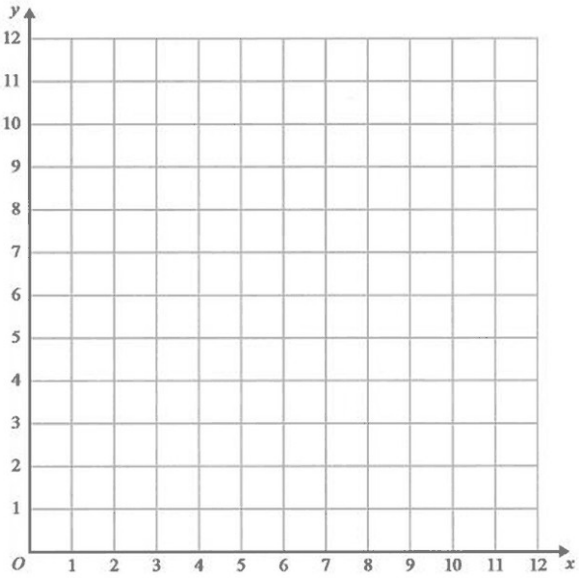
(2)

Question 2 (3 marks)

On the grid, show by shading the region defined by the inequalities

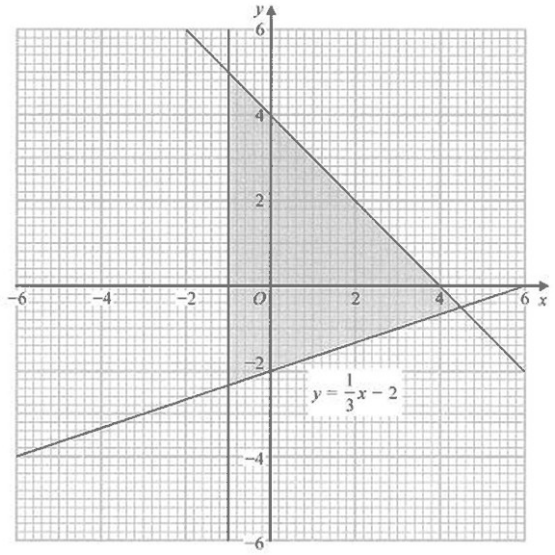
$y > 5$ and $y < 2x + 1$ and $x + y < 10$

Label your region **R**.



Question 3 (3 marks)

The shaded region in the diagram is bounded by three lines.
The equation of one of the lines is given.



Write down the three inequalities that define the shaded region.

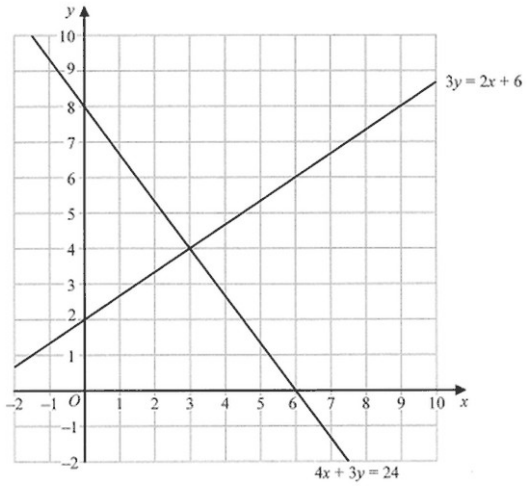
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Question 4 (1 mark)

The diagram shows two straight lines drawn on a grid.



(a) Write down the solution of the simultaneous equations

$$3y = 2x + 6$$
$$4x + 3y = 24$$

$x =$ .....

$y =$ .....

(1)

(b) Show, by shading on the grid, the region defined by all five of the inequalities

$x \geqslant 0$ $y \geqslant 0$ $x + y \geqslant 4$ $3y \leqslant 2x + 6$ $4x + 3y \leqslant 24$

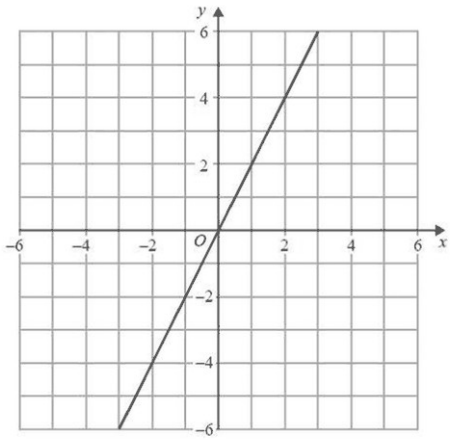
Label the region **R**.

Question 5 (5 marks)

The line with equation $y = 2x$ is drawn on the grid.

(a) On the same grid, draw the line with equation $4x + 3y = 12$

(2)

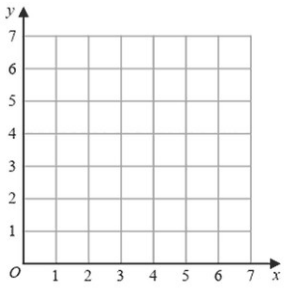


(b) Show, by shading on the grid, the region defined by all four inequalities

$$y \leqslant 2x$$
$$4x + 3y \leqslant 12$$
$$y \geqslant -3$$
$$x \leqslant 4$$

(3)

Question 6 (4 marks)



(a) On the grid, draw and **label** the straight line with equation

(i) $x = 1.5$

(ii) $y = x$

(iii) $x + y = 6$

(3)

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$x \geqslant 1.5$ $y \geqslant x$ $x + y \leqslant 6$

Label the region **R**.

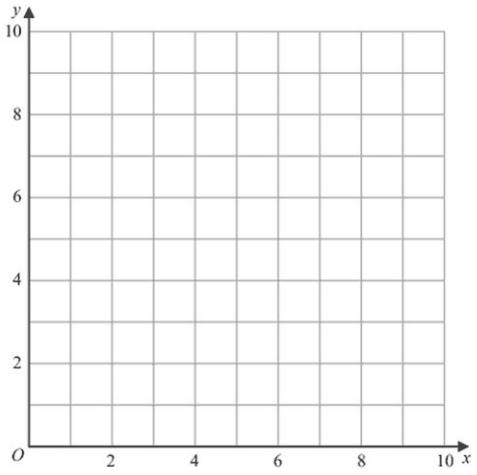
(1)

Question 7 (3 marks)

Show, by shading on the grid, the region defined by **all three** of the inequalities

$x \leqslant 6$ $y \geqslant 2$ $y \leqslant x + 1$

Label the region **R**

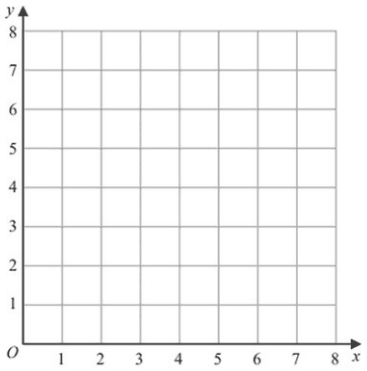


(3)

Question 8 (4 marks)

(a) On the grid, draw and label with its equation the straight line with equation

(i) $y = 1$ (ii) $x = 2$ (iii) $x + y = 7$



(3)

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$y \geqslant 1$ $x \geqslant 2$ $x + y \leqslant 7$

Label the region **R**.

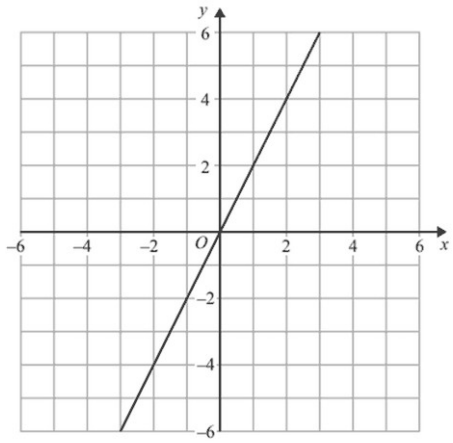
(1)

Question 9 (5 marks)

The line with equation $y = 2x$ is drawn on the grid.

(a) On the same grid, draw the line with equation $4x + 3y = 12$

(2)



(b) Show, by shading on the grid, the region defined by all four inequalities

$y \leqslant 2x$

$4x + 3y \leqslant 12$

$y \geqslant -3$

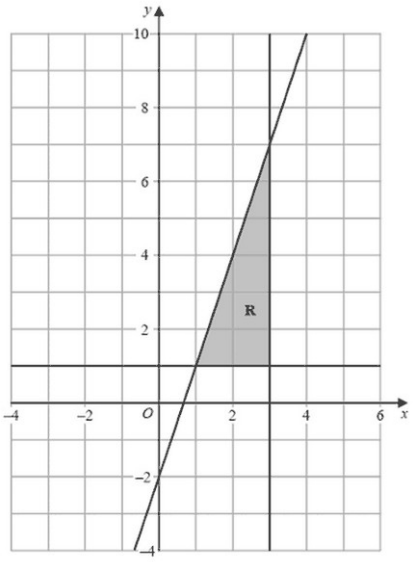
$x \leqslant 4$

(3)

Question 10 (3 marks)

The shaded region **R**, shown in the diagram below, is bounded by the straight line with equation $y = 3x - 2$ and by two other straight lines.

Write down the three inequalities that define region **R**.

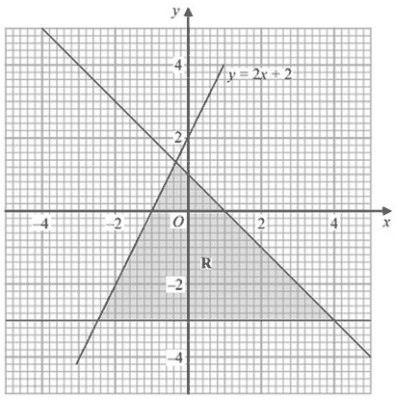


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Question 11 (3 marks)



The region **R**, shown shaded in the diagram, is bounded by three straight lines.

Write down the three inequalities that define **R**.

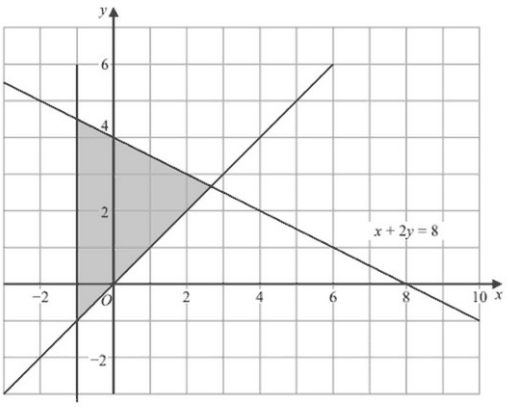
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Question 12 (3 marks)

The shaded region in the diagram is bounded by three lines.
The equation of one of the lines is given.



Write down three inequalities that define the shaded region.

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